

DATA ANALYTICS PROJECT

E-Commerce Sales & Customer Behavior Analysis

*A relational data analysis using SQL and Python — from raw transactional
data to business recommendations*

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SQL (SQLite) · Python (pandas, matplotlib) · Relational Data Modeling
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OVERVIEW

Executive Summary

This report analyzes 12,000 e-commerce orders across 3,000 customers and 250 products, using SQL for data extraction and aggregation and Python for statistical analysis and visualization. The objective was to answer eight core business questions spanning revenue trends, regional performance, product profitability, customer retention, delivery performance, and payment behavior.

Four findings stand out for their operational relevance. Revenue grew consistently across the two-year window with no signs of seasonal collapse. A base of repeat customers — under a third of all customers — generates roughly 92% of total revenue. Delivery times beyond 14 days are associated with a measurable, consistent drop in review scores. And Electronics is the single largest revenue-driving category despite not leading in unit volume, a pattern with direct implications for inventory and marketing prioritization.

2,378,850 TOTAL REVENUE (JOD)	12,000 ORDERS ANALYZED	92% REVENUE FROM REPEAT BUYERS	4.46 / 5 RATING, FAST DELIVERY
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SECTION 1

1 Methodology

Data Source

This project uses a synthetically generated e-commerce dataset structured to mirror real-world platforms such as Olist (Brazilian E-Commerce), covering customers, products, orders, order items, payments, and reviews. The dataset was generated with realistic statistical properties — skewed category popularity, gamma-distributed delivery times, a defined repeat-customer segment, and delivery-linked review score degradation — so the analysis demonstrates genuine data relationships rather than random noise.

Relational Data Model

The schema consists of six linked tables, joined primarily through order_id and customer_id:

customers customer_id (PK) customer_city signup_date	orders order_id (PK) customer_id (FK) order_date, status delivery_days	products product_id (PK) category unit_price, weight_g
payments order_id (FK) payment_type installments payment_value	order_items order_item_id (PK) order_id (FK) product_id (FK) quantity, price	reviews order_id (FK) review_score

Tools & Workflow

- SQLite — relational database engine used to store and query the dataset
- SQL — eight analytical queries using joins, common table expressions, aggregation, and conditional logic
- Python (pandas) — executing queries and shaping results into structured DataFrames
- Python (matplotlib) — all chart generation, styled to one consistent visual system throughout this report

SECTION 2

Findings

Eight business questions, answered end-to-end: SQL query, result, key insights, and a recommendation.

FINDING 2.1

2.1 Monthly Revenue Trend

Business Question *How has revenue evolved month over month, and is growth consistent?*

SQL Query

```
SQL
SELECT
  strftime('%Y-%m', o.order_date) AS month,
  ROUND(SUM(p.payment_value), 2) AS total_revenue,
  COUNT(DISTINCT o.order_id) AS order_count,
  ROUND(SUM(p.payment_value) / COUNT(DISTINCT o.order_id), 2) AS avg_order_value
FROM orders o
JOIN payments p ON o.order_id = p.order_id
WHERE o.status = 'delivered'
GROUP BY month
ORDER BY month;
```

Result

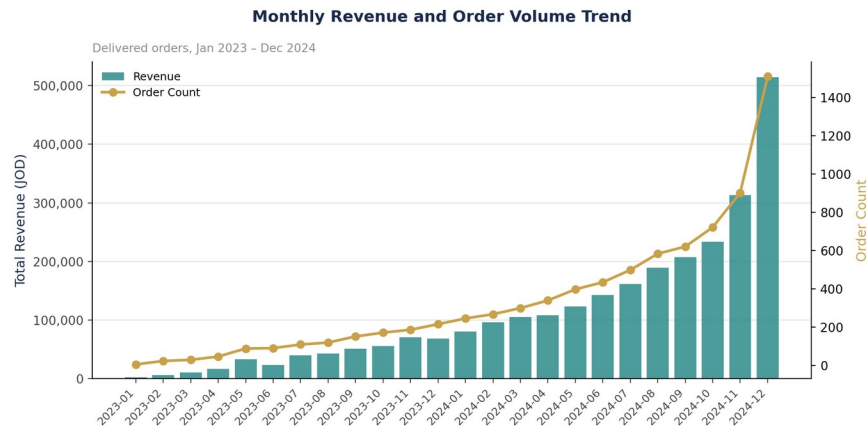


Figure 2.1 — Monthly revenue (bars, left axis) and order count (line, right axis), Jan 2023–Dec 2024.

Key Insights

- 01 Revenue grew month over month through 2023 and into 2024, with no extended periods of decline.
- 02 Order count tracks closely with revenue, indicating growth is driven by volume rather than isolated large orders.
- 03 Average order value stayed within a stable band (~JOD 300–380), suggesting consistent basket composition rather than one-off high-value spikes.

RECOMMENDATION

Revenue growth is broad-based rather than dependent on a few large orders, which is a healthier growth pattern to report to stakeholders and easier to forecast against.

FINDING 2.2

2.2 Revenue by City

Business Question Which regions generate the most revenue, and where should regional resources be prioritized?

SQL Query

SQL

```
SELECT
  c.customer_city,
  COUNT(DISTINCT o.order_id) AS total_orders,
  ROUND(SUM(p.payment_value), 2) AS total_revenue,
  ROUND(AVG(p.payment_value), 2) AS avg_order_value
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
JOIN payments p ON o.order_id = p.order_id
WHERE o.status = 'delivered'
GROUP BY c.customer_city
ORDER BY total_revenue DESC;
```

Result

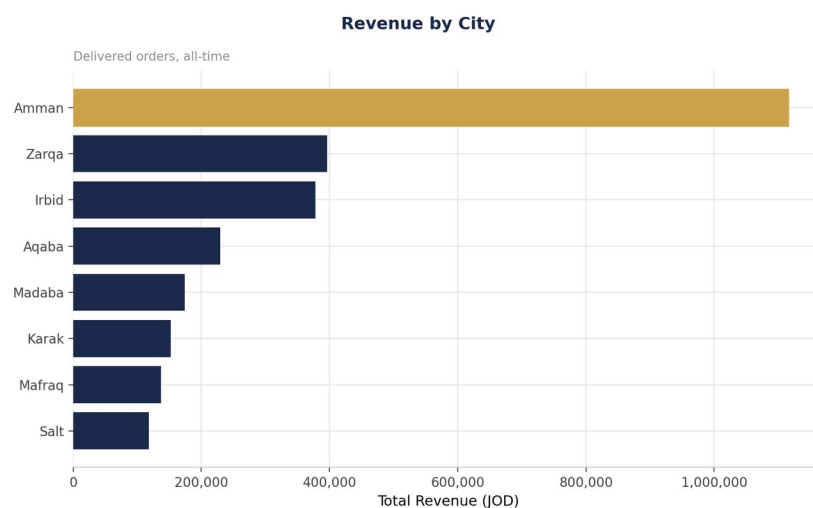


Figure 2.2 — Total revenue by customer city, delivered orders only. Highlighted bar marks the top-performing city.

Key Insights

- 01** Amman accounts for the largest share of both orders and revenue, consistent with its population density.
- 02** Zarqa and Irbid follow as the second tier, each contributing roughly a third of Amman's volume.
- 03** Average order value is fairly uniform across cities (JOD 303–350), meaning regional revenue differences are driven by order volume, not basket size.

RECOMMENDATION

Marketing and logistics investment should scale with order volume by city rather than assuming uniform demand — Amman, Zarqa, and Irbid together represent the majority of revenue.

FINDING 2.3

2.3 Revenue by Product Category

Business Question Which product categories should the business prioritize in inventory and marketing?

SQL Query

SQL

```
SELECT
    pr.category,
    COUNT(oi.order_item_id) AS units_sold,
    ROUND(SUM(oi.price * oi.quantity), 2) AS category_revenue,
    ROUND(AVG(oi.price), 2) AS avg_unit_price
FROM order_items oi
JOIN products pr ON oi.product_id = pr.product_id
JOIN orders o ON oi.order_id = o.order_id
WHERE o.status = 'delivered'
GROUP BY pr.category
ORDER BY category_revenue DESC;
```

Result

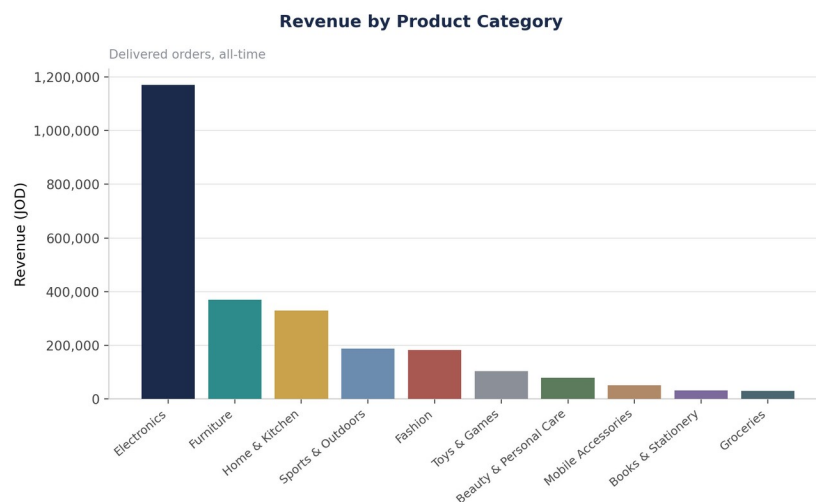


Figure 2.3 — Total revenue by product category, delivered orders only.

Key Insights

- 01** Electronics is the top revenue category by a wide margin, despite Home & Kitchen and Fashion selling more units — driven by a much higher average unit price.
- 02** Furniture, though low in unit volume, ranks second in revenue for the same reason: high price point per item.
- 03** Groceries and Books & Stationery sell reasonably high volumes but contribute the least revenue, reflecting low margins per unit.

RECOMMENDATION

Revenue-per-category should guide promotional budget allocation over unit volume alone — Electronics and Furniture justify disproportionate marketing spend relative to their share of items sold.

FINDING 2.4

2.4 Repeat vs. One-Time Customers

Business Question *What share of customers are repeat buyers, and how much more do they spend over their lifetime?*

SQL Query

SQL

```
WITH customer_orders AS (
  SELECT o.customer_id,
         COUNT(DISTINCT o.order_id) AS n_orders,
         SUM(p.payment_value) AS total_spent
  FROM orders o
  JOIN payments p ON o.order_id = p.order_id
  WHERE o.status = 'delivered'
  GROUP BY o.customer_id
)
SELECT
  CASE WHEN n_orders > 1 THEN 'Repeat Customer'
       ELSE 'One-Time Customer' END AS customer_type,
  COUNT(*) AS num_customers,
  ROUND(AVG(total_spent), 2) AS avg_lifetime_spend,
  ROUND(SUM(total_spent), 2) AS total_revenue_contribution
FROM customer_orders
GROUP BY customer_type;
```

Result

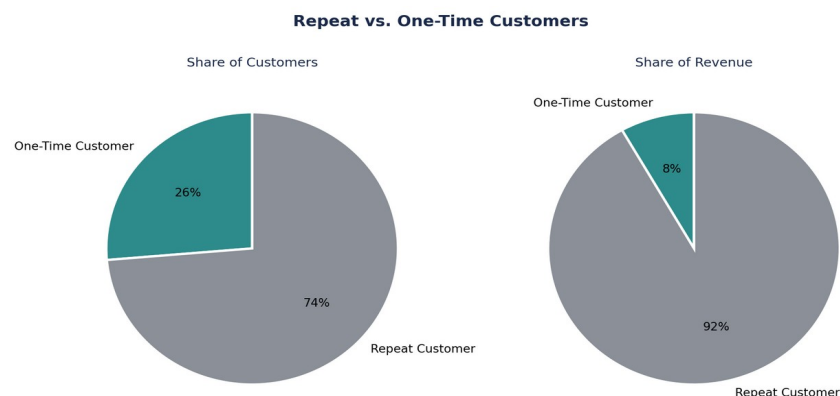


Figure 2.4 — Share of customers (left) versus share of revenue (right), split by repeat vs. one-time buyers.

Key Insights

- 01** Repeat customers make up a minority of the customer base but contribute the overwhelming majority of total revenue.
- 02** Average lifetime spend for repeat customers is several multiples higher than one-time buyers.
- 03** This concentration means customer retention has outsized leverage on total revenue compared to new customer acquisition alone.

RECOMMENDATION

Retention-focused initiatives — loyalty programs, post-purchase engagement, personalized offers — likely offer a higher return than acquisition spend alone, given how much revenue is concentrated in repeat buyers.

FINDING 2.5

2.5 Delivery Speed vs. Customer Satisfaction

Business Question Does slower delivery measurably hurt customer satisfaction, as reflected in review scores?

SQL Query

SQL

```
SELECT
  CASE
    WHEN o.delivery_days <= 3 THEN '1-3 days'
    WHEN o.delivery_days <= 7 THEN '4-7 days'
    WHEN o.delivery_days <= 14 THEN '8-14 days'
    ELSE '15+ days'
  END AS delivery_bucket,
  COUNT(*) AS num_orders,
  ROUND(AVG(r.review_score), 2) AS avg_review_score
FROM orders o
JOIN reviews r ON o.order_id = r.order_id
WHERE o.status = 'delivered' AND o.delivery_days IS NOT NULL
GROUP BY delivery_bucket
ORDER BY MIN(o.delivery_days);
```

Result

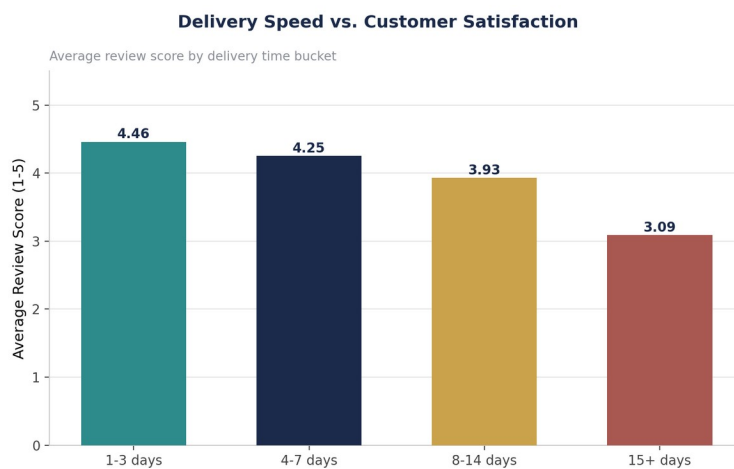


Figure 2.5 — Average review score (1-5) by delivery time bucket.

Key Insights

- 01** There is a clear, consistent decline in average review score as delivery time increases.
- 02** Orders delivered in 1-3 days average 4.46/5, while orders taking 15+ days drop to 3.09/5 — a decline of over one full point.
- 03** The steepest drop occurs beyond the 14-day mark, suggesting this is a meaningful threshold for customer tolerance.

RECOMMENDATION

Delivery time should be treated as a direct driver of customer satisfaction, not just an operational metric. Setting an internal SLA target under 14 days — and flagging orders approaching that threshold — could materially protect review scores and repeat-purchase likelihood.

FINDING 2.6

2.6 Payment Method Preferences

Business Question *How do customers prefer to pay, and does payment method correlate with order value?*

SQL Query

```
SQL
SELECT
  payment_type,
  COUNT(*) AS num_orders,
  ROUND(AVG(payment_value), 2) AS avg_order_value,
  ROUND(SUM(payment_value), 2) AS total_value
FROM payments
GROUP BY payment_type
ORDER BY total_value DESC;
```

Result

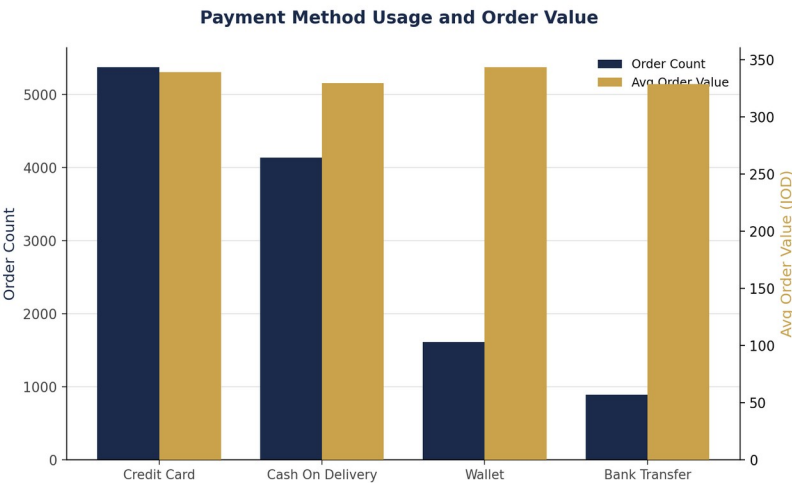


Figure 2.6 — Order count (bars, left axis) and average order value (bars, right axis) by payment method.

Key Insights

- 01** Credit card is the most-used payment method, accounting for the largest share of both order count and total value.
- 02** Cash on delivery remains significant (34% of orders), indicating a meaningful segment of customers who prefer to pay upon receipt.
- 03** Average order value is broadly similar across payment types, suggesting payment method is a preference choice rather than a constraint tied to basket size.

RECOMMENDATION

Cash on delivery should remain a supported option given its meaningful share, even as the business may want to incentivize card payments — for example with small discounts — to reduce delivery-time cash-handling logistics.

FINDING 2.7

2.7 Order Cancellation Rate by City

Business Question *Are order cancellations concentrated in specific regions, pointing to a localized fulfillment issue?*

SQL Query

SQL

```
SELECT
    c.customer_city,
    COUNT(*) AS total_orders,
    SUM(CASE WHEN o.status = 'canceled' THEN 1 ELSE 0 END) AS canceled_orders,
    ROUND(100.0 * SUM(CASE WHEN o.status = 'canceled' THEN 1 ELSE 0 END)
          / COUNT(*), 2) AS cancellation_rate_pct
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.customer_city
ORDER BY cancellation_rate_pct DESC;
```

Result

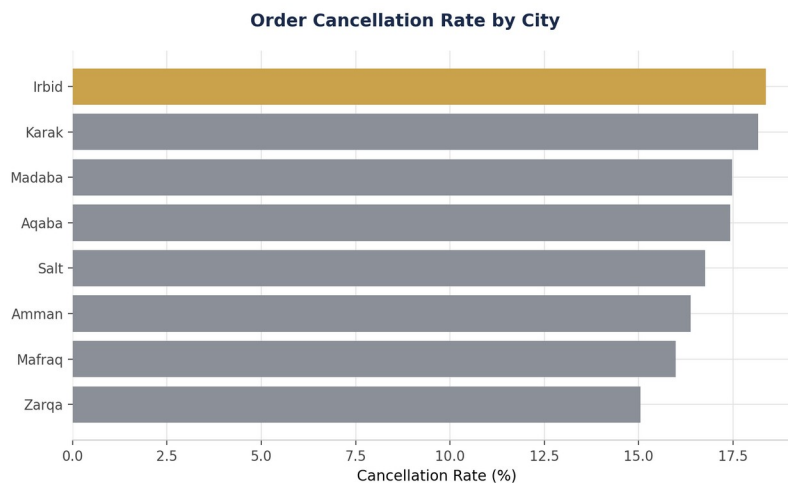


Figure 2.7 — Cancellation rate (%) by customer city. Highlighted bar marks the highest rate.

Key Insights

- 01** Cancellation rates range narrowly between roughly 15% and 18% across all cities.
- 02** Irbid and Karak show the highest cancellation rates, though the spread across cities is modest rather than a sharp outlier.
- 03** Because the variation is narrow, cancellations appear to reflect a platform-wide pattern rather than a single region's fulfillment problem.

RECOMMENDATION

Given the narrow spread, root-cause investigation into cancellations — payment failures, stock issues, delivery estimate mismatches — should be pursued platform-wide rather than targeted at one city.

FINDING 2.8

2.8 Top 10 Products by Revenue

Business Question Which individual SKUs are the biggest revenue drivers?

SQL Query

```
SQL
SELECT
  oi.product_id,
  pr.category,
  ROUND(SUM(oi.price * oi.quantity), 2) AS product_revenue,
  SUM(oi.quantity) AS units_sold
FROM order_items oi
JOIN products pr ON oi.product_id = pr.product_id
JOIN orders o ON oi.order_id = o.order_id
WHERE o.status = 'delivered'
GROUP BY oi.product_id
ORDER BY product_revenue DESC
LIMIT 10;
```

Result

PRODUCT ID	CATEGORY	REVENUE (JOD)	UNITS SOLD
129	Electronics	92,694.24	108
27	Electronics	86,927.46	102
220	Electronics	67,655.86	101
250	Electronics	65,298.24	112
102	Electronics	64,942.95	95
206	Electronics	64,651.00	85
46	Electronics	62,008.32	96
23	Electronics	61,689.04	98
212	Electronics	57,598.62	93
75	Electronics	55,952.99	101

Key Insights

- 01 All ten top-revenue products belong to the Electronics category, reinforcing the category-level finding in Section 2.3.
- 02 Revenue is fairly evenly spread across these ten products rather than dominated by a single "hero" SKU — the top product (92,694 JOD) is only about 1.7x the tenth (55,953 JOD).

RECOMMENDATION

Electronics inventory depth and supplier relationships for these top ten SKUs should be prioritized to avoid stockouts, given their outsized contribution relative to the rest of the 250-product catalog.

SECTION 3

3 Conclusion & Recommendations

Across all eight analyses, three themes stand out as the highest-priority takeaways for the business:

- 01 Retention over acquisition — repeat customers, under a third of the customer base, generate the large majority of revenue. Retention programs likely offer stronger ROI than acquisition-only strategies.
- 02 Delivery speed is a satisfaction lever, not just an operations metric — the clear decline in review scores as delivery time increases suggests an internal SLA (e.g. under 14 days) would have a direct, measurable effect on customer satisfaction and repeat-purchase behavior.
- 03 Revenue concentration by category and product — Electronics and Furniture punch above their unit-volume weight in revenue terms. Inventory, marketing, and supplier-relationship investment should be weighted accordingly rather than spread evenly across the catalog.

Skills Demonstrated

- SQL — multi-table joins, common table expressions (CTEs), conditional aggregation (CASE WHEN), grouped analytics
- Python — pandas for data manipulation, matplotlib for custom-styled visualization, an end-to-end pipeline from raw data to chart
- Data modeling — relational schema design across six linked tables
- Business analysis — translating raw query output into prioritized, actionable recommendations

SECTION 4

4 Appendix: Data Dictionary

Reference for every field used across the six source tables.

TABLE	FIELD	DESCRIPTION
customers	customer_id	Unique customer identifier (primary key)
customers	customer_city	City of residence
customers	signup_date	Date the customer registered
products	product_id	Unique product identifier (primary key)
products	category	Product category
products	unit_price	List price per unit (JOD)
orders	order_id	Unique order identifier (primary key)
orders	status	delivered, shipped, or canceled
orders	delivery_days	Days between order and delivery
order_items	quantity	Units of the product in this order
payments	payment_type	credit_card, cash_on_delivery, wallet, bank_transfer
reviews	review_score	Customer rating, 1 (lowest) to 5 (highest)